called the producer's surplus. It is represented in the following diagram.

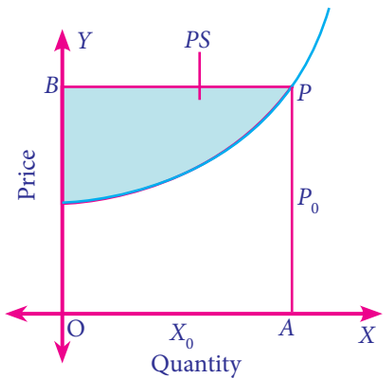


Fig. 3.17

Mathematically, producer's surplus (PS) can be defined as,

PS = (Area of the rectangle OAPB) − (Area below the supply function from $x = 0$ to $x = x_0$)

$$PS = x_0 p_0 - \int_0^{x_0} g(x)dx$$

Example 3.28

Find the producer's surplus defined by the supply curve $g(x) = 4x+8$ when $x_0 = 5$.

Solution:

$g(x) = 4x + 8$ and $x_0 = 5$

$p_0 = 4(5) + 8 = 28$

$$PS = x_0 p_0 - \int_0^{x_0} g(x)dx$$

$$= (5 \times 28) - \int_0^5 (4x+8)dx$$

$$= 140 - \left[4\left(\frac{x^2}{2}\right) + 8x\right]_0^5$$

$$= 140 - (50 + 40)$$

$$= 50 \text{ units}$$

Hence the producer's surplus = 50 units.

Example 3.29

The demand and supply function of a commodity are $p_d = 18 - 2x - x^2$ and $p_s = 2x - 3$. Find the consumer's surplus and producer's surplus at equilibrium price.



The point of intersection of demand and supply curves is called equilibrium point.
At equilibrium point $q_d = q_s$

Solution:

Given $P_d = 18 - 2x - x^2$; $P_s = 2x - 3$

We know that at equilibrium prices $p_d = p_s$

$18 - 2x - x^2 = 2x - 3$

$x^2 + 4x - 21 = 0$

$(x-3)(x+7) = 0$

$x = -7$ or 3

The value of x cannot be negative, $x = 3$

When $x_0 = 3$

$\therefore \quad p_0 = 18 - 2(3) - (3)^2 = 3$

$$CS = \int_0^x f(x)dx - x_0 p_0$$

$$= \int_0^3 (18 - 2x - x^2)dx - 3 \times 3$$

$$= \left[18x - x^{2-} \frac{x^3}{3}\right]_0^3 - 9$$

$$= 18(3) - (3)^2 - \left(\frac{3^3}{3}\right) - 9$$

$CS = 27$ units

$$PS = x_0 P_0 - \int_0^x g(x)dx$$

$$= (3 \times 3) - \int_0^3 (2x-3)dx$$

$$= 9 - \left(x^2 - 3x\right)_0^3$$

$= 9$ units

Hence at equilibrium price,

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